

Tim Alamenciak

Curriculum Vitae

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EDUCATION

- 2024 Ph.D., Social and Ecological Sustainability, University of Waterloo
- 2012 M.A., Journalism, Toronto Metropolitan University
- 2010 B.A. (Hons.), Peace and Conflict Studies, University of Waterloo

PROFESSIONAL APPOINTMENTS

- 2024– Postdoctoral Fellow, Bennett Lab, Department of Biology, Carleton University, Ottawa, Ontario
- 2024– Adjunct Assistant Professor, School of Environment, Resources and Sustainability, University of Waterloo, Waterloo, Ontario
- 2024– Research Fellow, DRAGEN Lab, St. Jerome's University, Waterloo, Ontario
- 2024– Senior Research Scientist, Restoration Futures Lab, University of Victoria, Victoria, British Columbia

VISITING POSITIONS AND RESIDENCIES

- 2024–2025 Visiting Researcher, Centre for Interdisciplinary Research (ZiF), Bielefeld University, Germany. Research group: "Mapping Evidence to Theory in Ecology," invited by Dr. Tina Heger.

PUBLICATIONS

Refereed Journal Articles

- Ryan Y. Hodgson, Steven A. Robinson, Amelie C. Boutin, Felix K. Chan, Joseph R. Bennett, Rachel T. Buxton, J. Harry Caufield, Dalal E.L. Hanna, **Tim Alamenciak**. "Assessing the Effectiveness of Ontology-Grounded AI Term Extraction Using OntoGPT for Environmental Evidence Synthesis." *Environmental Evidence*.
- Tim Alamenciak**, Nancy Shackelford, Logan Rehberg, Ash Baron, Steven D. Murphy, Eric Higgs, Tina Heger, Alina Fisher, Bruno Travassos-Britto, Ryan Stephenson. "Identifying practitioner and researcher collaboration needs to improve ecosystem restoration in Canada." *Socio-Ecological Practice Research*.

Jeffrey O. Hanson, Jenny L. McCune, **Tim Alamenciak**, Joseph R. Bennett. "Increasing the credibility of conservation plans through citizen science." *Biological Conservation*.

Josie South, Roxana Barbulescu, Rafael L. Macedo, Camille L. Musseau, Simone Guareschi, **Tim Alamenciak**, et al.. "Parallels between biological invasions and human migration are flawed and undermine both disciplines. Response to Ahmed et al." *BioScience*.

Tim Alamenciak, Elise Gornish, Stephen D. Murphy. "Dimensions of Effective Volunteer Restoration Techniques in North America." *Restoration Ecology*. [Audio]

Tim Alamenciak, Stephen D. Murphy. "What makes a convivial community tool? Investigating grassroots ecological restoration." *Ecology & Society*. [Audio]

Tim Alamenciak, Stephen D. Murphy. "Motivations for Volunteers to Participate in Ecological Restoration: A Systematic Map." *Restoration Ecology*. [Audio]

Tim Alamenciak, Dorian Pomezanski, Nancy Shackelford, Stephen D. Murphy, Steven J. Cooke, Line Rochefort, Sonia Voicescu, Eric Higgs. "Ecological Restoration Research in Canada: Who, What, Where, When, Why, and How?." *FACETS*. [Audio]

William J. Sutherland, Jake M. Robinson, David C. Aldridge, **Tim Alamenciak**, Matthew Armes, Nina Baranduin, Andrew J. Bladon, et al.. "Creating Testable Questions in Practical Conservation." *Conservation Evidence Journal*.

Anita Lazurko, **Tim Alamenciak**, Lowine Stella Hill, Ella-Kari Muhl, Augustine Kwame Osei, Dorian Pomezanski, Kyle Schang, Dilruba Fatima Sharmin. "What Will a PhD Look Like in the Future?." *World Futures Review*.

Manuscripts in Submission or Preparation

Courtney D. Robichaud, Christine Beaudoin, **Tim Alamenciak**, Jaimie Vincent, Steven J. Cooke, Vivian M. Nguyen, Richard Schuster, Nathan Young, Joseph R. Bennett. "Data, politics, and funding cause uncertainty for conservation practitioners." *Conservation Science and Practice* (Major revisions).

Tim Alamenciak, Amy Bachhuber, Joseph R. Bennett, Steven J. Cooke, Kian L. Drew, Daniel Dylewsky, Emily McKnight, Pat Moore, Ana Hernandez Martinez De La Riva, Bronwyn Rayfield. "Machines in the loop: Challenges and opportunities for environmental evidence synthesis research in the artificial intelligence era." *FACETS* (Under review).

Federica Bocchi, Aline Potiron, Eleonore Slabbert, **Tim Alamenciak**, Anika Grose, Birgitta Konig-Ries, Lotte Korell, Carlos Santana, Tina Heger. "Operationalizing the CARE principles in evidence synthesis for ecology and conservation." *Nature Communications* (Under review).

Tim Alamenciak, Stephen D. Murphy. “Which Aspects of Restoration Project Organization Affect Volunteer Engagement in Community-Based Initiatives?.” *Restoration Ecology* (Under review).

Steven J. Cooke, Kevin A. Adeli, Trina Rytwinski, Andrew N. Kadykalo, Jennifer Provencher, Vivian Nguyen, Joseph Bennett, Christina Davy, Rachel Buxton, Dalal Hanna, Jesse C. Vermaire, Sean Landsman, Nathan Young, Graeme Auld, Danika Littlechild, Jennifer M. Holzer, Meagan Harper, Andrew Howarth, **Tim Alamenciak**, Lauren Lawson, Jayme Lewthwaite, Erin E. Stukenholtz, Paul A. Smith, Ilona Naujokaitis-Lewis, Josie Hughes, Barbara Frei, Amanda Martin, Amie Black, Richard Pither, Douglas MacNearney, Kristen Lalla, Carmen Galan-Acedo and Christopher Cvitanovic. “Supporting frontline workers in the biodiversity crisis by empowering and enabling practitioners to embrace conservation evidence.” *Socio-Ecological Practice Research* (Major revisions).

Conference Proceedings

Tim Alamenciak, Carlos Alberto Arnillas, J. Harry Caufield, Katherine Compton, Kian Drew, Robert Fruehstueckl, Tina Heger, Birgitta Konig-Ries, Chris Mungall, Sierra Moxon, Justin Reese, Jordan Tardif, Lars Vogt. “Ecolink: Towards a Knowledge Graph Schema for Complex Environmental Systems.” *New Trends in Theory and Practice of Digital Libraries*.

Preprints

Lars Vogt, Birgitta Konig-Ries, **Tim Alamenciak**, Joshua I. Brian, Carlos Alberto Arnillas, Lotte Korell, Robert Fruehstueckl, Tina Heger. “A Framework for FAIR and CLEAR Ecological Data and Knowledge.” *arXiv*.

AWARDS AND HONORS

2025	PSAC 77000 Postdoctoral Fellow Award - Finalist
2022	Certificate in University Teaching Award - \$1,000

GRANTS AND FELLOWSHIPS

2026	DFG Workshop Fund for September EcoWeaver workshop - €22,000
2026	Environmental Data Science Innovation & Impact Lab (ESIIL) 2026 working group - CMSMapper: Optimal climate mitigation planning for nature and people - \$65,000 USD (co-PI)
2026	NVIDIA Academic Grant Program Award, NVIDIA DGX Spark - \$5,000 (PI)
2025	St. Jerome's University Faculty Research Grant (EcoTransform Hackathon) - \$10,000 (co-PI with Kian Drew)
2025	PSAC 77000 Professional Development Fund - \$2,000 (PI)

- 2024 NSERC Alliance fund, "Evaluating the potential of temporary conserved areas for biodiversity conservation" with Ducks Unlimited Canada - \$355,000 (co-PI with Dr. Joseph Bennett)
- 2022 University of Waterloo Sustainability Action Fund - \$20,000 (co-PI with Dr. Shefaza Esmail)
- 2021 University of Waterloo Sustainability Action Fund - \$20,000 (co-PI with Dr. Shefaza Esmail)
- 2020 Ontario Trillium Fund, ecological restoration education with Ignatius Jesuit Centre - \$70,000 (co-applicant)
- 2020 Waterloo Environment Students Endowment Fund - \$1,000 (PI)
- 2019 Waterloo Environment Students Endowment Fund - \$300 (PI)

INVITED TALKS

- 2025 “Three stories of restoration and conservation engagement..” School of Environment, Resources and Sustainability Salon; Carleton BIOL 5534; Knowledge Mobilization Community of Practice; University of Waterloo PSCI 303.
- 2025 “Toolkit for Restoration Ecology Knowledge (TReK)..” AI Colloquium, Lamarr Institute, Dortmund, Germany.
- 2024 “How to recruit and retain more volunteers..” Webinar for Carolinian Canada.
- 2024 “Fuzzy cognitive mapping using knowledge graphs for restoration ecology..” Interdisciplinary Perspectives on Knowledge Synthesis in Ecology workshop, University of Bielefeld.
- 2024 “Mapping evidence to theory in ecology: Addressing the challenges of generalization and causality..” Residency at Centre for Interdisciplinary Research (ZiF), Bielefeld University.
- 2024 “People-powered ecosystem restoration..” University of Arizona, Gornish Lab.
- 2024 “Taming Complexity in Ecology..” Workshop, Germany.
- 2024 “Lowering barriers, changing behaviours..” Guest lecture (Invited by Dr. Christina Davy).
- 2024 “Volunteer engagement in ecological restoration..” Guest lecture: ENVS 195 – Introduction to Environmental Studies.
- 2022 “Advocacy 101: How to talk about climate change with politicians..” ClimateActionWR sector committee talk.
- 2022 “Guest speaker..” Landscapes of Holistic Sustainability and Wellbeing Retreat, Wilfrid Laurier University.
- 2022 “Championing Change in the Field and Office..” Designing Change for a Living Planet – World Wildlife Fund.

- 2022 “Comparing community-led tree planting programs..” University of Waterloo, ERS 253.
- 2021 “Working Together and Giving Back: Building and sustaining volunteer motivation..” Wholistic Restoration – Ignatius Jesuit Centre.
- 2021 “Decolonizing Worldviews: Unpacking nature and wilderness..” Wholistic Restoration – Ignatius Jesuit Centre.
- 2020 “Who gets to care for nature? Participation in ecological restoration..” Guest Lecture – University of Waterloo PACS 310.
- 2020 “Assessment in restoration ecology..” University of Waterloo, ERS 342.

CONFERENCE ACTIVITY

Papers Presented

- 2026 “The enduring benefits of short-term conservation agreements.” Canadian Society for Ecology and Evolution Annual Conference.
- 2025 “Ecolink: Towards a Knowledge Graph Schema for Complex Environmental Systems..” EcoDL Workshop at the Theory and Practice of Digital Libraries conference, Tampere, Finland.
- 2024 “Boots & Suits: Mobilizing knowledge in ecosystem restoration..” Society for Ecological Restoration North American Conference, Vancouver.
- 2024 “Engaging instructors and students in transformative learning amid polycrisis..” University of Waterloo Teaching and Learning Conference.
- 2023 “Volunteer engagement in ecological restoration..” Environment Graduate Students Association conference, Waterloo, Ontario.
- 2023 “Ecological Restoration Research in Canada: Who, What, Where, When, Why and How?.” RE3 Conference: Reclaim, Restore, Rewild, Quebec City, Quebec.
- 2021 “How to design techniques for volunteer ecological restoration..” National Extension Conference on Volunteerism-Lite.
- 2021 “Analyzing Canadian ecological restoration literature with bibliometric analysis and a systematic map..” Evidence Synthesis & Meta-Analysis in R Conference.
- 2021 “Charting the evidence landscape: A systematic map and bibliometric analysis of ecological restoration in Canada..” Society for Ecological Restoration Conference.

TEACHING EXPERIENCE

University of Waterloo

Sustainability: The Future We Want (Spring 2026)

Communities and Sustainability (sessional lecturer, Winter 2024) [in-person]

Ecosystem Assessment (teaching assistant, Spring 2021) [online]

Translational Ecology (teaching assistant, Winter 2021) [online]

Ecological Restoration (teaching assistant, Fall 2020) [online]

Introduction to Environment Studies; Advanced Topics in Ecological Restoration; Place, Identity and Community (teaching assistant, 2019-2020)

Carleton University

Honours Thesis Supervision, Bachelor of Science (Fall 2025; Winter 2026)

Advances in Applied Ecology (group mentor, Fall 2024)

Honours Thesis Supervision, Bachelor of Science (Spring 2024)

RESEARCH EXPERIENCE

2020–2023 Research Associate, Ecological Restoration Knowledge Synthesis Project, University of Victoria (in collaboration with University of Waterloo, Carleton University, and Laval). PI: Dr. Eric Higgs. Multi-year team effort conducting the first systematic map of ecological restoration literature in Canada.

2021–2022 Project Manager, Campus EcoMap, University of Waterloo. Secured \$20,000 in funding; supervised five undergraduate research associates; produced ArcGIS/QGIS index of ecological research on campus.

SERVICE TO THE PROFESSION

Manuscript Review

Restoration Ecology; Sustainability Science; Society & Natural Resources; Ecology & Society; People & Nature

Grant Review

Digital Research Alliance - EDIA Champions Program; Earth Science Information Partners (ESIP) - Community Resilience RFP

Leadership

Board Member at-large, Environmental Studies Association of Canada (ESAC), 2024 - Current

Board Member, Emerging Leaders for Biodiversity (ELB), 2024 - 2025

Board Member, Society for Ecological Restoration (Ontario), 2020 - 2021

DEPARTMENTAL AND UNIVERSITY SERVICE

Member, Knowledge Mobilization Community of Practice (UW), 2025 - Current

Member, Natural Spaces Working Group (UW), 2025 - Current

Land Stewardship Committee Member, Ignatius Jesuit Centre, 2024 - 2025

Writing Cafe Organizer, Weekend Warriors Writing Cafe, University of Waterloo, 2021 - 2022

Member, Ecological and Environmental Advisory Committee, Region of Waterloo, 2019 - 2021

NON-ACADEMIC WORK

2021–2023 Communications Manager, Reep Green Solutions

2015–2018 Digital Media Producer, TVO

2012–2015 Staff Reporter, Toronto Star